pd-trigger-lite — Design Document

PCB-0016-A

hwde v1 pipeline

generated 2026-09-25 18:09:54

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1 Board and Run Metadata

Field	Value
board	pd-trigger-lite
part number	PCB-0016-A
workspace	`boards/PCB-0016-A_pd-trigger-lite`
phase	P8
gate status	all 5 recorded gates pass
state created	2026-09-24T02:31:06
state updated	2026-09-24T03:47:38

2 Overview

Brief: pd-trigger-lite

Owner's words: "a simple pd trigger board like the showcase one ... No need to copy the existing ... Ideally low cost ... the zips and component placement files so I can order PCB cheaply from JLC".

Design intent (set by the dispatching planning session; unattended run - the owner is away, every recipe question is decided here and recorded with state.py decision):

- USB-C PD SINK trigger, 2-layer, 1 oz, small (aim <= ~25 x 15 mm; bigger only if DRC needs it).
- CH224K (LCSC C970725) with its minimal application circuit.

- Voltage selectable after assembly with no tool beyond a soldering iron (solder jumpers / 0R links), sensible assembled default 12 V.
- Output: VBUS + GND on two large 2.54 mm-compatible through-hole pads, no screw terminal. Rated 3 A (no e-marker, so 3 A is the cable/source ceiling).
- One power LED if it costs one basic resistor + one basic LED.
- A bulk/input cap per datasheet. No TVS/fuse unless the datasheet requires it.
- JLC BASIC parts everywhere possible; USB-C connector and CH224K are extended - cheapest widely stocked power-only receptacle exposing CC1/CC2.
- Deliverables: gerber zip, JLC BOM.csv + CPL.csv, order quote for qty 5 with assembly, owner guide PDF, README. Never place an order.
- Showcase boards/pd-trigger: learn from it, reuse its CH224K extraction, do not copy its layout or feature set.

3 Requirements

Requirements: pd-trigger-lite

Source: `brief/brief.md`. Written inline by the orchestrator (no requirements-analyst spawn: a one-block board, recorded as a decision). Unattended run: every open item below was answered by the orchestrator as the owner's delegate and recorded with `state.py decision`. Mode: none (the brief carries no mode token) - design normally.

1. Function

A low-cost USB-C Power Delivery trigger. A PD sink controller negotiates a fixed voltage from a USB-C PD charger and passes VBUS straight through to two output pads. No regulation, no conversion, no switching element in the power path. The requested voltage is chosen by solder links on the board: 9, 12, 15 or 20 V, shipped at 12 V.

2. Interfaces

- Input: USB-C receptacle, power-only use, PD SINK. CC1/CC2 to the controller; D+/D- to the controller's DP/DM per the datasheet reference (no USB data function).
- Output: VBUS and GND on two 2.54 mm-pitch plated through-holes (take a 1x2 header or wires). No screw terminal (cost).
- Voltage selection: one solder link per voltage (9/12/15/20 V), exactly one closed. The 12 V link ships fitted as a 0 ohm resistor; the others are open pads.

3. Power

- Input: USB-C PD source, 5 V default, negotiated 9/12/15/20 V.
- Output current rating: 3 A continuous at any voltage (a non-e-marked cable caps the contract at 3 A; the board makes no 5 A claim).
- Controller supply: from VBUS directly (CH224A VHV pin is rated 32 V).

4. Environment

Bench use, indoor, 0-50 C ambient. No enclosure, no battery, no mains.

5. Size & mounting

Target <= 25 x 15 mm (soft; grow only if DRC needs it). No mounting holes (cost/size; the board hangs off its cable and output wires).

6. Quantity & budget

Quantity 5, JLCPCB PCB + economy PCBA, lowest cost. JLC Basic parts wherever possible.

7. Assembly

Single-sided SMT at JLC. The 1x2 output header is optional and hand-fitted by the owner.

8. Compliance/safety flags

- >3 A: NO (3 A max). Mains: NO. Battery: NO.
- Protection: no TVS/fuse - the controller datasheet requires none (decision recorded).

9. Open questions

None open. Every question was answered by the orchestrator as delegate (see state.json decisions): controller part, selection method, default voltage, output format, current rating, LED, bulk cap, protection.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Unattended run: orchestrator answers every owner question as delegate; H checkpoints self-approved and recorded	Owner is away and answers nothing (dispatch brief).
P1	P1 research: no research agents spawned; the CH224 manual V2.1 (showcase parts/C970725_CH224K.pdf covers CH224A) is in hand and read directly	one-block board, cost cap in dispatch brief
P2	D1 controller CH224A C42459160 (not CH224K): VHV 32 V direct from VBUS, Rset table published; removes dropper, sense R and pull-ups	architecture/decisions.md (delegate, unattended)
P2	D2 voltage select = Rset mode, one resistor+link per voltage (6.8k/24k/56k/120k = 9/12/15/20 V), 12 V link fitted as 0R	architecture/decisions.md (delegate, unattended)
P2	D3 J1 = 6-pin power-only USB-C C2798175 (20 V/3 A rated); DP/DM unconnected (PD on CC only)	architecture/decisions.md (delegate, unattended)

Phase	What	Why
P2	D4 no TVS/fuse/bulk cap: datasheet requires only 1 uF on VHV; brief says none unless required (risk disclosed in guide)	architecture/decisions.md (delegate, unattended)
P2	D5 output = two unassembled 2.54 mm plated holes (board feature)	architecture/decisions.md (delegate, unattended)
P2	D6 power LED from VBUS via 10k (PG sink current unpublished)	architecture/decisions.md (delegate, unattended)
P2	D7 2L 1.6 mm 1 oz, VBUS 3 A at dt 20 C, B.Cu GND pour	architecture/decisions.md (delegate, unattended)

Sheet plan

pd-trigger-lite - sheet plan

One flat root sheet `kicad/pd-trigger-lite.kicad_sch`, pwr_base 1, generator `kicad/gen/root.py`. 14 symbols, one chain, every cross-block net is a rail (VBUS, GND) or a two-part stub - hierarchy would buy nothing.

Canonical nets: `VBUS`, `GND` (power symbols, bare); `/CC1`, `/CC2`, `/CFG1`, `/SEL9`, `/SEL12`, `/SEL15`, `/SEL20` (Rset-to-link midpoints), `/LED_A`.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`.

5 Schematic

Gate `erc` (phase P4): **pass** after 5 attempt(s); last 2026-09-24T03:10:15, failing 0 of 0.

Review waivers

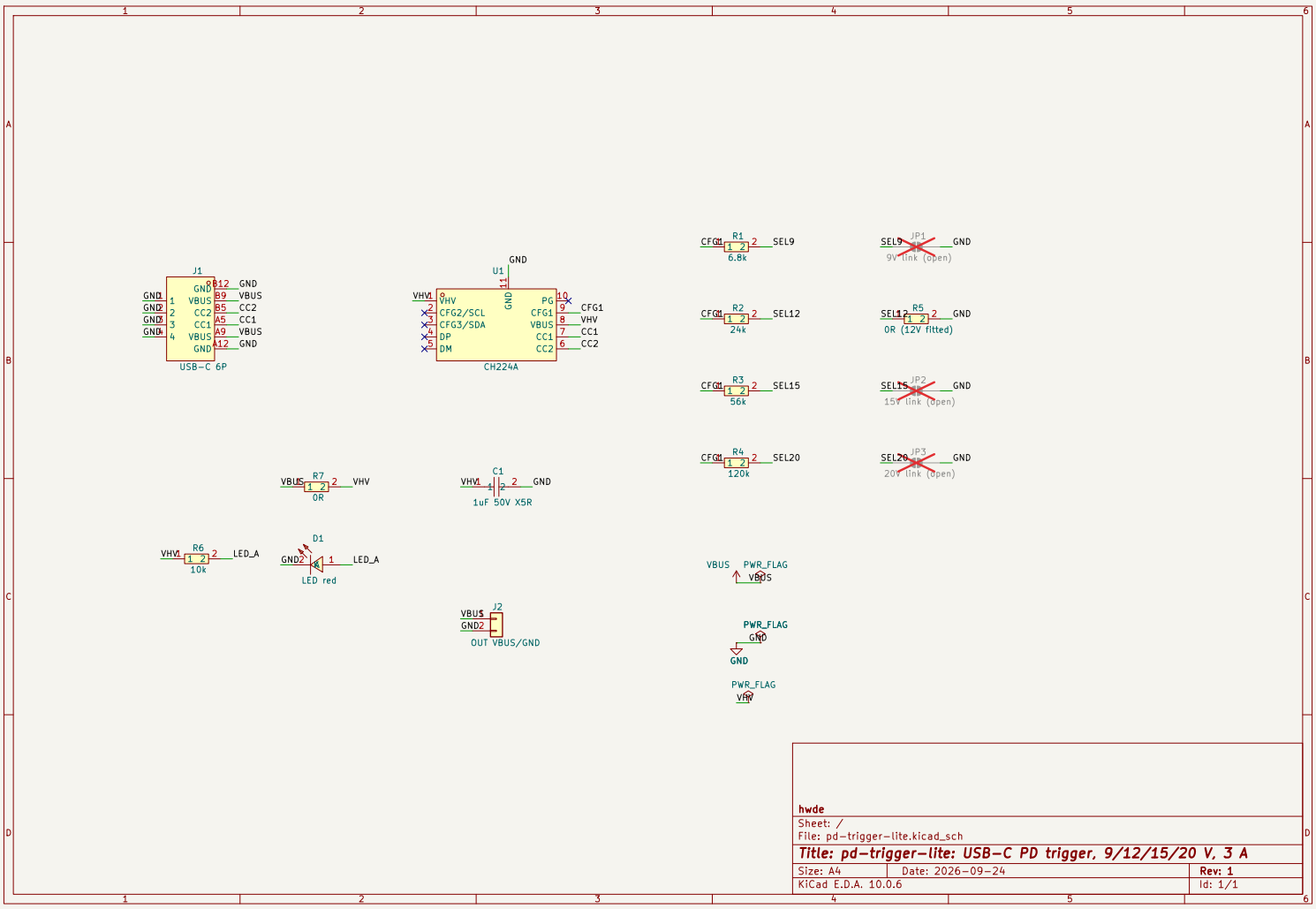
ERC / schematic-review waivers (pd-trigger-lite, 2026-09-24, delegate)

ERC: 0 errors, 0 warnings - nothing waived. Schematic review (reports/review-schematic.md), 0 errors, 2 warnings, both accepted:

- W1 no interlock for "exactly one link": zero links floats CFG1, two links parallel two Rset values. Accepted: the board ships with one link (R5) fitted; the guide and silk labels say move the one 0R. All links are now identical 0603 lands, so "relocate the 0R" is the natural rework.
- W2 no TVS/fuse/bulk against 20 V hot-plug ringing: decision D4 (datasheet requires none, brief says none). Disclosed in the guide.

Post-review edits (not re-reviewed, ERC re-run 0/0): R7 0R VBUS->/VHV split (width rule vs 1 mm pitch), JP1-3 moved from solder-jumper footprints to empty 0603 lands (DNP).

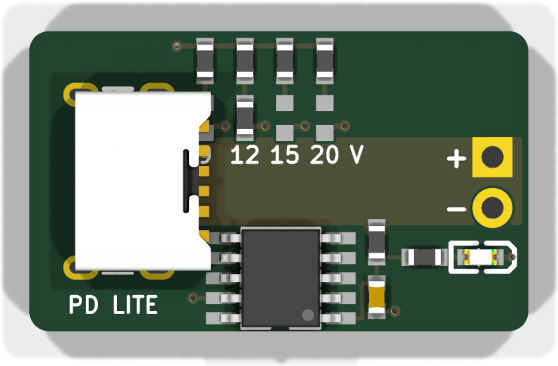
The full schematic PDF follows.



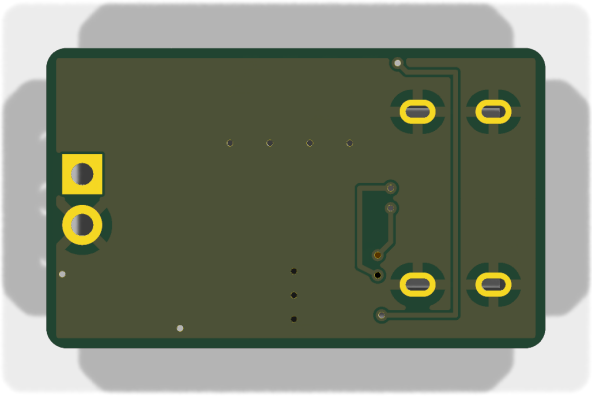
hwde	
Sheet: /	
File: pd-trigger-lite.kicad_sch	
Title: pd-trigger-lite: USB-C PD trigger, 9/12/15/20 V, 3 A	
Size: A4	Date: 2026-09-24
KiCad E.D.A. 10.0.6	Id: 1/1
	Rev: 1

6 Layout

Gate `place` (phase P6): **pass** after 3 attempt(s); last 2026-09-24T03:22:45, failing 0 of 0.
Gate `drc_routed` (phase P7): **pass** after 2 attempt(s); last 2026-09-24T03:23:53, failing 0 of 0.



`reports/pd-trigger-lite_top.png`



`reports/pd-trigger-lite_bottom.png`

7 Verification

Gate `verify` (phase P8): **pass** after 2 attempt(s); last 2026-09-24T03:23:58, failing 0 of 17.

Verification checks

Check	Status	Findings	Note
`check_return_path`	pass	0	
`check_current`	violations	17	
`check_decoupling`	pass	0	
`check_diffpair`	pass	0	
`check_creepage`	pass	0	
`check_thermal`	pass	0	
`check_silk`	pass	0	
`check_pdn`	pass	0	

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0 *(no digest recorded)*

P1 *(no digest recorded)*

P2

```
P1+P2 (inline, 2026-09-24):  CH224A PD sink (VHV direct from VBUS), Rset
voltage select with
one link per voltage (12 V 0R fitted), 6P USB-C 20 V/3 A, VBUS LED, 2
unassembled output holes.
~10 assembled parts, 2 extended (U1, J1).  2L 1 oz, VBUS 3 A dt 20 C. Coverage
P2:  0 slots
(no diff pairs / topology blocks; the PD-sink interface reference was read
directly).
Riskiest decision:  no TVS (datasheet requires none; brief says none) -
disclosed.
```

P3 *(no digest recorded)*

P4

```
P4 (2026-09-24):  flat generated schematic (kicad/gen/root.py), 15 symbols; ERC
0/0; netlist_audit pass.
Reviewer (sonnet, fresh):  0 errors, 2 warnings (link interlock, no TVS)
accepted in reports/erc-waivers.md.
Post-review:  R7 0R split VBUS -> /VHV; JP1-3 now empty 0603 lands.  Needs
KICAD_SYMBOL_DIR on this host
and tools/kicad-extra (libwebkit2gtk-4.1 for the eeschema kiface).
```

P5 *(no digest recorded)*

P6 *(no digest recorded)*

P7

```
P5-P7 (2026-09-24):  board_init 25x15 r1, parity 0; rules_gen VBUS 1.248 mm
class.  Manual placement
(place_p6.ops.json):  J1 left edge mouth out, VBUS as F.Cu zone band y 5.3-9.7
to J2, Rset row top,
U1 + VHV/LED bottom.  Freerouting routed all signals on F.Cu straight through
the VBUS band (split the
zone, CFG1 under J1 on F.Cu), so it was discarded; hand-routed with route_edit
(route_edit_manual.json:
CC and CFG1 cross on B.Cu, 11 vias).  VBUS zone switched to solid pad
connection (3 starved thermals).
Refdes silk hidden, passive outlines cleared, labels added.  drc_routed PASS
0/0.
route_critical skipped:  KRT not installed; nothing critical to route (VBUS is
a plane).
```

P8 *(no digest recorded)*

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-09-24T02:55:20	erc	pass	0/0
2026-09-24T02:56:57	erc	pass	0/0
2026-09-24T02:59:48	erc	pass	0/0
2026-09-24T03:01:47	erc	pass	0/0
2026-09-24T03:10:15	erc	pass	0/0
2026-09-24T03:11:59	place	fail	5/5
2026-09-24T03:12:32	place	pass	0/0
2026-09-24T03:22:28	drc_routed	pass	0/0
2026-09-24T03:22:45	place	pass	0/0
2026-09-24T03:22:54	verify	fail	1/18
2026-09-24T03:23:53	drc_routed	pass	0/0
2026-09-24T03:23:58	verify	pass	0/17
2026-09-24T03:29:59	dfm	pass	0/0

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-09-24T02:48:45	delegate (owner away): blocks/parts/stackup per architecture/decisions.md; ballpark under 5 USD/board parts; coverage 0 slots; no TVS per brief
2	P4	approved	2026-09-24T03:10:43	delegate: ERC 0/0, review 0 err / 2 warn waived (reports/erc-waivers.md)
3	P6	approved	2026-09-24T03:22:46	delegate: top render reports/render_place, J1 mouth at left edge
4	P8	approved	2026-09-24T03:47:38	delegate: P8 verify review 0 err/0 warn; 17 check_current GND advisories waived (mA stubs, 0.8 mm J1 GND tracks 1.5 A each over 1.2 mm); VBUS pour >= 2.2 mm, GND B.Cu plane; CPL rot U1 90/J1 270/D1 180 verified; see reports/verify-review.md
5	P10	not reached		

Generation cost

Model cost of generating this board: **$24.26**.

Step	Cost
Before the workspace (brief and skill read)	$5.29
Brief and intake	$2.92
Architecture	$0.03
Schematic	$5.21
Placement	$0.51

Step	Cost
Routing	$2.24
Verification	$5.38
DFM, order files and guide	$2.67

Guide rebuilds by later skill-wide rounds (board renders with parts, fresh guide renders, this cost record) are skill work and are not counted; the fresh-guide-renders round left no transcript on this host.

The worker loop logged $9.37. 1 of its iterations timed out, and a timed-out iteration logs no cost; the session transcripts hold the whole total.

10 Artifact Index

File	Size	Note
state.json	20,296 B	
kicad/PCB-0016-A_pd-trigger-lite.kicad_pcb	87,619 B	
kicad/PCB-0016-A_pd-trigger-lite.kicad_sch	73,989 B	
fab/PCB-0016-A_pd-trigger-lite_gerbers.zip	14,727 B	
fab/BOM.csv	368 B	
fab/CPL.csv	398 B	
brief/brief.md	1,387 B	
requirements.md	2,340 B	
architecture/blocks.md	1,734 B	
architecture/constraints.json	1,260 B	
architecture/decisions.md	1,259 B	
architecture/sheets.md	400 B	
reports/board_init.json	7,392 B	
reports/bom_cpl.json	8,974 B	
reports/check_current.json	16,001 B	
reports/cost.json	4,503 B	
reports/drc_final.json	580 B	
reports/erc-waivers.md	800 B	
reports/erc_final.json	463 B	
reports/fab_export.json	2,293 B	
reports/gate-dfm.json	1,551 B	
reports/gate-drc_routed.json	1,060 B	
reports/gate-erc.json	861 B	
reports/gate-place.json	1,445 B	
reports/gate-verify.json	2,030 B	
reports/guide_facts.json	8,758 B	
reports/lib_pull.json	14,142 B	
reports/netlist_audit_p4.json	844 B	
reports/pd-trigger-lite_bottom.png	57,351 B	
reports/pd-trigger-lite_top.png	145,648 B	
reports/place_edit_p6.json	1,844 B	
reports/place_edit_silk.json	2,014 B	
reports/place_p6.ops.json	1,648 B	
reports/planes_gen.json	1,162 B	
reports/review-board.json	19 B	
reports/review-schematic.json	1,402 B	
reports/review-schematic.md	5,393 B	
reports/route_auto.json	46,348 B	
reports/route_edit_manual.json	9,164 B	
reports/route_edit_manual_result.json	6,150 B	
reports/rules_gen.json	1,028 B	
reports/schematic.pdf	84,474 B	

File	Size	Note
reports/silk_ops_manual.json	2,002 B	
reports/silk_place.json	20,413 B	
reports/verify-review.md	2,935 B	
reports/verify_all.json	17,758 B	